

Name:

Stud.ID:

Vision and Cognitive Systems

A.Y. 2022/2023

Midterm Exam - December 12, 2022

Exam Instructions

- Write your *name* and *Stud.ID (matricola)* on **every** page.
- You have just one page (one side) to answer a question.
- You will have **exactly** 75 min to complete your exam.

Question 1 (2 points)

Please provide an example of edge detector (i.e. the kernel matrix) and illustrate what are the main analogies with biological vision (if any), and the applications in which edge detection could be a fundamental component.

Question 2 (3 points)

Describe in details a bag-of-visual-words pipeline highlighting all the main steps of computation. Finally, you should briefly introduce how it can be extended in order to better represent the spatial elements depicted in a given image.

Question 3 (3 points)

Describe all the major components of the SIFT algorithm and provide some examples of applications in which SIFT features are the key ingredient.

Question 4 (2 points)

Please provide a clear definition for a Gaussian filter (i.e. the kernel mask) and introduce another filter that can give you a similar output. You should also describe the main differences between the two.